Technische Hochschule Rosenheim

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Electrical Engineering

Development and Validation of a Concept for Modular Battery Management System

Master Thesis

by

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Declaration of Originality

I declare that I have authored this thesis independently, that I have not used other than the declared sources / resources, and that I have explicitly marked all material which has been quoted either literally or by content from the used sources.

Rosenheim, 15.06.2026

Yara Hany

Abstract

In english

Contents

1 Introduction

1.1 Background and Motivation

The transition towards sustainable transportation and electrified mobility has led to a significant increase in the use of Lithium-ion (Li-ion) battery systems. These batteries are widely utilized in applications such as electric vehicles, unmanned aerial vehicles, and electric Vertical Take-Off and Landing (eVTOL) aircraft due to their high energy density, low self-discharge rate, and long cycle life. As battery systems continue to increase in size and complexity, ensuring their safe and reliable operation becomes increasingly important. Li-ion batteries are sensitive to operating conditions such as overvoltage, undervoltage, overcurrent, and excessive temperatures. If these conditions are not properly monitored and controlled, battery degradation, reduced performance, or safety-critical failures may occur.

To address these challenges, Battery Management Systems (BMS) are employed to monitor, protect, and manage battery packs during operation. A BMS is responsible for acquiring battery measurements, supervising safety limits, estimating battery states, and communicating relevant information to other system components. Consequently, the BMS represents a critical subsystem in modern battery-powered applications.

In industrial applications, battery systems often need to be scalable and adaptable to different configurations. Conventional BMS architectures may require significant modifications when battery capacity or system requirements change. Therefore, modular BMS concepts have gained increasing attention due to their flexibility, scalability, and potential for reuse across different battery systems. Furthermore, the development of embedded control systems increasingly follows a Model-Based Design (MBD) approach. This methodology enables the implementation, simulation, and automatic code generation of control algorithms within a unified development environment. To ensure the correctness of the developed software throughout the development process, several validation stages such as Model-in-the-Loop (MIL), Software-in-the-Loop (SIL), Processor-in-the-Loop (PIL), and Hardware-in-the-Loop (HIL) are commonly applied.

This thesis focuses on the development and validation of a concept for a modular Battery Management System using a Model-Based Design workflow. The proposed concept is implemented in MATLAB/Simulink and evaluated through MIL, SIL, PIL, and BIL validation stages to assess its functionality and consistency throughout the development process.

1.2 Problem Statement

The increasing complexity of battery-powered systems requires Battery Management Systems that are scalable, flexible, and reliable. Conventional BMS architectures often face limitations when adapting

to different battery configurations and system requirements. A modular BMS architecture offers the potential to improve scalability, maintainability, and reusability while supporting different battery system configurations.
The proposed concept aims to provide a reusable BMS architecture in which battery-specific parameters can be adapted to different battery configurations without requiring fundamental changes to the implemented control logic.

1.3 Objectives

The objective of this thesis is to develop and validate a modular Battery Management System architecture based on a scalable arrangement of Analog Front End (AFE) modules and a central STM32 microcontroller. The proposed concept shall enable monitoring of battery packs with varying numbers of cells through the addition of AFE modules while maintaining a common software architecture. Furthermore, the concept is evaluated using a Model-Based Design workflow and validated through MIL, SIL, PIL, and HIL testing stages.

1.4 Scope of the Thesis

The developed BMS concept includes a battery plant model and a Stateflow-based BMS logic layer. The BMS monitors battery voltage, current, and temperature signals, evaluates operating conditions, and generates status and fault indications. The generated outputs represent information intended for communication with higher-level control systems such as a flight controller. The implementation of the flight controller and communication network is outside the scope of this thesis.

1.5 Methodology

This thesis follows a Model-Based Design (MBD) approach for the development and validation of a modular Battery Management System (BMS). The development process begins with the definition of system requirements and the design of the overall architecture. The proposed system consists of a battery plant model and a BMS logic layer implemented in MATLAB/Simulink and Stateflow.
The battery plant model is used to simulate the electrical behavior of the battery pack and provide measurements such as cell voltages, pack current, and temperature values. These signals are processed by the BMS logic, which evaluates the battery state, detects abnormal operating conditions, and generates status and fault indications.
Following the model development phase, the implemented functionality is verified through a sequence of validation stages. First, Model-in-the-Loop (MIL) testing is performed to verify the behavior of the Simulink model. Subsequently, Software-in-the-Loop (SIL) testing is conducted to compare the behavior of the generated code with the original model. Processor-in-the-Loop (PIL) testing is then carried out using the target STM32F446RE microcontroller to evaluate the execution of the generated code on the target processor. Finally, Battery-in-the-Loop (BIL) testing is performed to assess the system under realistic battery operating conditions.

The results obtained from the different validation stages are compared to evaluate the consistency of the Model-Based Design workflow and to verify the functionality of the proposed modular BMS concept.

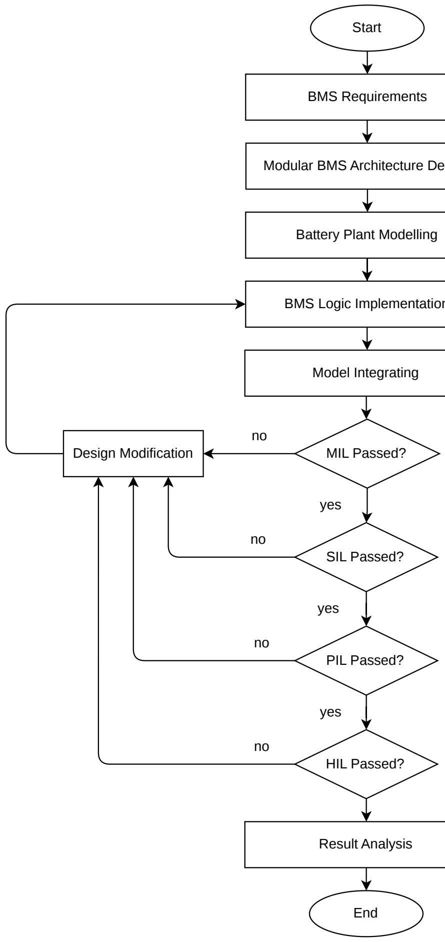

Figure 1.1 Methodology followed in this thesis.

1.6 Thesis Structure

This thesis is organized into seven chapters.
Chapter 1 introduces the research topic and presents the motivation, problem statement, objectives, scope of work, and methodology of the thesis.
Chapter 2 provides the theoretical background and literature review related to battery technologies, Battery Management Systems (BMS), modular BMS architectures, and Model-Based Design (MBD). Furthermore, the concepts of Model-in-the-Loop (MIL), Software-in-the-Loop (SIL), Processor-in-the-Loop (PIL), and Battery-in-the-Loop (BIL) validation are discussed.
Chapter 3 presents the system requirements and the proposed modular BMS architecture. The hardware components and their integration within the overall system are described.
Chapter 4 describes the development and implementation of the battery plant model and the BMS logic using MATLAB/Simulink and Stateflow.
Chapter 5 presents the validation methodology and the test procedures applied during MIL, SIL, PIL, and BIL validation.
Chapter 6 discusses the obtained results and evaluates the performance of the developed modular BMS concept.
Chapter 7 concludes the thesis by summarizing the main findings and providing recommendations for future work.

2 Literature Review

2.1 Overview of Electric Vehicles

Electric vehicles (EVs) have gained significant attention in recent years as part of the global transition toward more sustainable transportation systems. Growing concerns regarding greenhouse gas emissions, climate change, and the shortage of fossil fuel resources have accelerated the adoption of electric propulsion technologies across various transportation sectors. In addition to environmental considerations, reducing dependence on fossil fuels has become an important strategic objective for many countries seeking to improve energy security and reduce exposure to fluctuations in global energy markets.

The increasing deployment of electric vehicles has created a growing demand for advanced energy storage systems capable of delivering high performance, reliability, and safety. Among the available energy storage technologies, lithium-ion batteries have emerged as the preferred choice due to their high energy density, high power density, low self-discharge rate, and relatively long cycle life. These characteristics make lithium-ion batteries suitable for a wide range of applications, including passenger vehicles, industrial systems, and unmanned aerial vehicles (UAVs).

Despite their advantages, lithium-ion batteries present several technical challenges. Battery performance and lifetime are highly dependent on operating conditions such as temperature, charging and discharging rates, and depth of discharge. Exposure to unfavorable operating conditions can accelerate battery degradation, reduce available capacity, increase internal resistance, and in extreme cases lead to safety hazards. Consequently, effective monitoring and management of lithium-ion batteries are essential to ensure safe operation, maximize performance, and extend battery lifetime [1].

Electric vehicles can generally be classified into three main categories: Battery Electric Vehicles (BEVs), Hybrid Electric Vehicles (HEVs), and Plug-in Hybrid Electric Vehicles (PHEVs). BEVs rely exclusively on electrical energy stored in batteries, whereas HEVs combine an electrical energy storage system with an additional power source, typically an internal combustion engine (ICE). PHEVs operate similarly to HEVs but can also be charged directly from an external electrical power source. Each architecture presents different requirements for battery sizing, energy management, and overall system design [2].

The platform considered in this thesis is a hybrid electric vertical takeoff and landing (eVTOL) unmanned aerial vehicle. In this system, the lithium-ion battery is responsible for supplying the high power demands required during vertical takeoff, hover, and landing operations. These flight phases place significant stress on the battery due to their high power requirements and rapidly changing load conditions. As a result, accurate battery monitoring and management are critical for ensuring operational safety, reliability, and efficiency. These challenges highlight the importance of Battery Management Systems (BMSs) and motivate the research presented in this thesis.

2.2 Unmanned Aerial Vehicles and Hybrid Drones

Unmanned Aerial Vehicles (UAVs) have experienced rapid growth in recent years due to their ability to perform a wide range of missions, including aerial surveillance, inspection, mapping, cargo transportation, and environmental monitoring. Depending on their aerodynamic configuration and flight characteristics, UAVs can be classified into fixed-wing, rotary-wing, and hybrid vertical takeoff and landing (VTOL) platforms.

Fixed-wing UAVs generate lift through aerodynamic surfaces and are generally characterized by high flight efficiency and long endurance. However, they require a runway or launching mechanism for takeoff and landing. In contrast, rotary-wing UAVs generate lift directly through rotating propellers, enabling vertical takeoff, landing, and hovering capabilities. Despite their operational flexibility, rotary-wing platforms typically exhibit lower flight endurance due to their continuous power requirements during flight.

Hybrid eVTOL UAVs combine the advantages of both fixed-wing and rotary-wing configurations. These platforms utilize electric lift motors during vertical takeoff and landing operations while relying on aerodynamic lift generated by wings during cruise flight. As a result, hybrid eVTOL systems can achieve both vertical mobility and improved flight endurance, making them suitable for long-range and heavy-payload applications.

Fully electric UAVs rely exclusively on batteries as their energy source. While this approach offers a simple architecture and eliminates direct emissions during operation, the achievable flight endurance is constrained by the energy density of current battery technologies. Increasing mission duration typically requires larger battery packs, which increase the overall vehicle mass. Since aircraft performance is highly sensitive to weight, battery mass becomes a critical design consideration, particularly for larger UAV platforms [3].

To overcome these limitations, hybrid propulsion systems have been developed by combining batteries with additional energy sources such as fuel cells or internal combustion engines. In such systems, the battery is responsible for supplying high peak power during demanding flight phases, including takeoff, hover, transition, and landing. The secondary energy source provides the average power required during cruise flight, thereby extending the overall mission endurance [2].

The power demand of a hybrid eVTOL UAV varies significantly throughout the flight mission. Vertical takeoff and landing phases require high power output to generate sufficient thrust to support the aircraft weight. During transition and cruise flight, the power requirements are typically reduced as aerodynamic lift generated by the wings contributes to supporting the vehicle. Consequently, the battery system is subjected to highly dynamic load profiles and must be capable of delivering large current levels while maintaining safe and reliable operation.

Hybrid propulsion architectures offer several advantages for UAV applications. The combination of a high-energy source, such as a fuel cell, and a high-power battery system allows each technology to operate within its optimal range. In addition, the battery can act as an energy buffer, compensating for transient power demands and fluctuations in the primary energy source. This improves system reliability, enhances flight endurance, and reduces stress on the propulsion system components. As a result, effective battery management becomes essential for ensuring the safety, performance, and longevity of the energy storage system in hybrid eVTOL platforms [3].

2.3 Lithium-Ion Battery Technology

Lithium-ion (Li-ion) batteries have become the dominant energy storage technology for electric vehicles, UAVs, and portable electronic devices due to their high energy density, high power density, and relatively long cycle life. Compared to alternative battery technologies such as lead-acid and nickel-metal hydride batteries, Li-ion cells provide superior performance while maintaining a relatively low weight, making them particularly suitable for aerospace applications where weight is a critical design constraint [4].

2.3.1 Fundamentals of Lithium-Ion Batteries

A lithium-ion battery consists of a positive electrode (cathode), a negative electrode (anode), an electrolyte, and a separator. The battery chemistry is generally identified according to the cathode material used. Common lithium-ion chemistries include Lithium Cobalt Oxide (LCO), Lithium Nickel Cobalt Aluminum Oxide (NCA), Lithium Nickel Manganese Cobalt Oxide (NMC), and Lithium Iron Phosphate (LFP). The anode is typically composed of graphite, although alternative materials are being investigated to improve battery performance.

The nominal voltage of a lithium-ion cell is typically around 3.6–3.7 V. Depending on the chemistry and manufacturer specifications, the maximum charging voltage is generally 4.2 V, while the minimum discharge voltage ranges between 2.5 V and 3.0 V. Lithium-ion cells are manufactured in various form factors, including cylindrical, prismatic, and pouch cells, each offering different advantages in terms of packaging, thermal management, and energy density[5].

2.3.2 Cell Structure and Operating Principle

The cathode and anode are separated by a porous separator soaked with electrolyte, which allows lithium ions to move between the electrodes while preventing electrical short circuits. During discharge, lithium ions move from the anode to the cathode through the electrolyte, while electrons flow through the external circuit, supplying power to the load. During charging, an external power source forces lithium ions to move back from the cathode to the anode, restoring the stored energy within the cell. Battery capacity is commonly expressed in ampere-hours (Ah), representing the amount of charge that can be delivered over time. The charge and discharge current are often expressed using the C-rate, where a 1C discharge corresponds to discharging the full battery capacity in one hour [6].

2.3.3 Charging and Discharging Mechanisms

Lithium-ion batteries are typically charged using a Constant Current–Constant Voltage (CC–CV) charging method. During the constant current phase, the battery is charged at a predefined current, commonly around 1C. Once the cell voltage reaches its maximum charging voltage, the charger transitions to the constant voltage phase, during which the charging current gradually decreases until a predefined termination current is reached [7]. The CC–CV charging profile is illustrated in Figure 2.1.

Fast charging can be achieved using several techniques, as illustrated in Figure 2.2. Although fast charging significantly reduces charging time, it may accelerate battery degradation, except in battery-

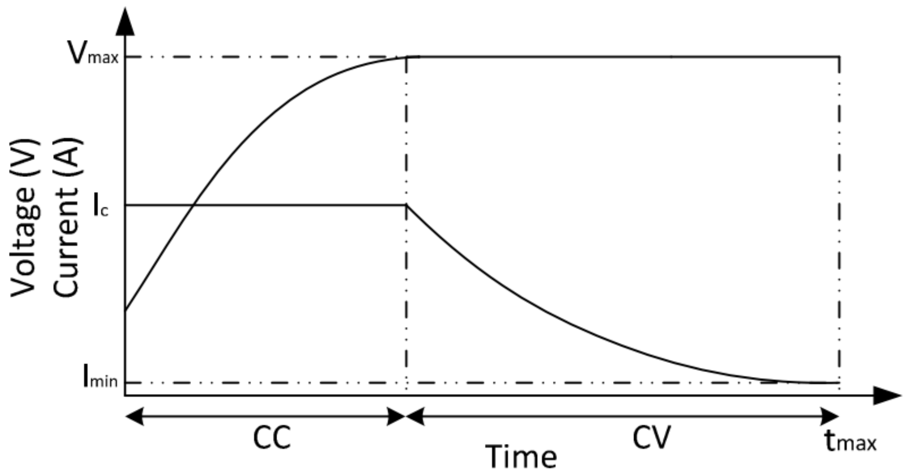

Figure 2.1 CC-CV Charging Technique.
Source: Reproduced from [7].

swapping approaches, due to increased temperature and electrochemical stress. Similarly, excessive discharge currents can generate significant heat, reducing battery lifetime and potentially creating unsafe operating conditions. In extreme cases, operation outside the recommended voltage, current, or temperature limits may result in thermal runaway [8].

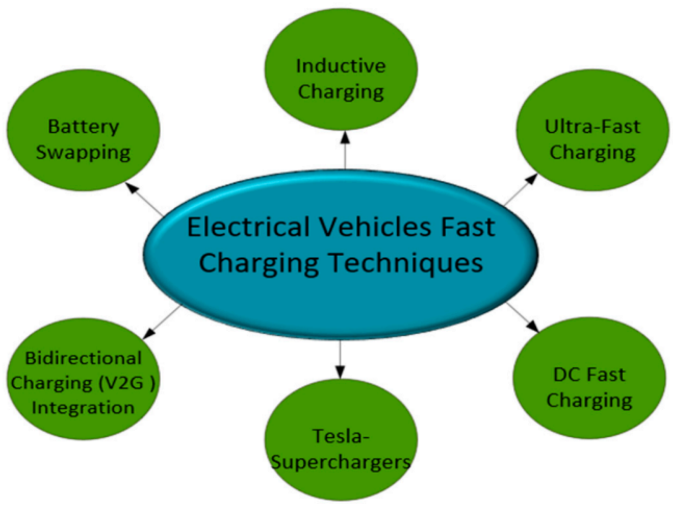

Figure 2.2 EV Fast charging techniques.
Source: Reproduced from [8].

In addition to operational discharge, lithium-ion batteries also experience self-discharge, which results in a gradual loss of stored energy even when the battery is not connected to a load. For long-term storage, batteries are typically maintained at a moderate state of charge rather than being stored fully charged[7].

2.3.4 Battery Performance Characteristics

Several parameters are commonly used to evaluate lithium-ion battery performance. Energy density, often expressed in Wh/kg, describes the amount of energy stored per unit mass and is one of the most important characteristics for aerospace applications. Modern lithium-ion batteries typically achieve energy densities between 200 and 350 Wh/kg, with ongoing research focused on increasing these values through advanced materials and solid-state battery technologies.
Power density, expressed in W/kg, represents the ability of the battery to deliver power relative to its mass. High power density is particularly important for UAV applications, where large current demands occur during takeoff, landing, and maneuvering operations.
Battery lifetime is commonly evaluated through cycle life and State of Health (SOH). The cycle life represents the number of charge-discharge cycles a battery can complete before its usable capacity falls below a specified threshold, commonly 80% of its initial capacity. SOH estimation remains an active area of research due to the complexity of battery degradation mechanisms and their dependence on operating conditions [5].

2.3.5 Safety Considerations and Battery Management

Despite their advantages, lithium-ion batteries are sensitive to overcharging, overdischarging, excessive current, and elevated temperatures. These conditions can accelerate degradation, reduce battery performance, and in severe cases lead to thermal runaway. Consequently, Battery Management Systems (BMSs) are employed to continuously monitor battery voltage, current, temperature, and operating conditions. The BMS plays a critical role in ensuring safe operation, maximizing battery lifetime, and maintaining reliable system performance [5].

2.4 Battery Pack Architectures

2.4.1 Series and Parallel Cell Confuguration

Lithium-ion battery packs are typically constructed by connecting individual cells in series, parallel, or a combination of both configurations to satisfy the voltage and power requirements of a specific application. Cells connected in series increase the overall battery voltage while maintaining the same capacity as a single cell. In contrast, cells connected in parallel increase the overall capacity and current capability of the battery pack while maintaining the same voltage level. Therefore, the selection of the battery configuration depends primarily on the electrical requirements of the application. Systems requiring high operating voltages generally utilize a larger number of series-connected cells, whereas applications with high current demands employ additional parallel cell groups to increase the available current capability.
In practical applications, battery packs are often designed using both series and parallel connections. This arrangement enables the battery system to achieve the desired voltage, capacity, and power output while optimizing weight and space constraints. For high-power applications such as electric vehicles and unmanned aerial vehicles, the battery architecture must be carefully designed to ensure that the electrical performance requirements are met without compromising safety, reliability, or

battery lifetime.

2.4.2 Battery Modules and Battery Packs

A battery pack is generally composed of multiple battery modules, where each module consists of a group of interconnected cells packaged as a single unit. The modular approach has become increasingly common in modern battery systems because it improves scalability, maintainability, and manufacturing flexibility. Instead of designing a unique battery pack for every application, a standardized battery module can be replicated and combined to achieve different voltage and capacity requirements. This approach reduces development effort and simplifies maintenance procedures since defective modules can be replaced individually without requiring replacement of the entire battery pack. The hierarchical relationship between battery cells, modules, and the complete battery pack is illustrated in Figure 2.3.

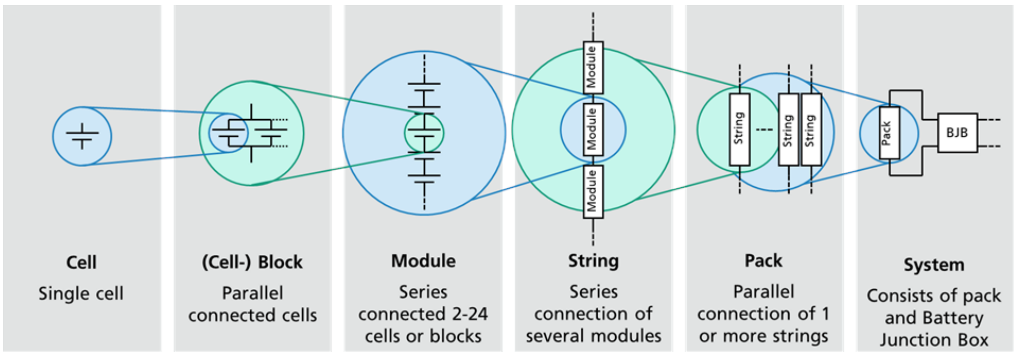

Figure 2.3 Naming conventions for battery system elements.
Source: Reproduced from [9].

The modular concept also extends to the Battery Management System. In a modular BMS architecture, each battery module is monitored by a dedicated Analog Front-End (AFE) device responsible for measuring cell voltages and temperatures. The number of monitored cells can be increased by adding additional AFE devices without requiring significant modifications to the software architecture. For example, some commercially available AFE devices are capable of monitoring up to fourteen battery cells simultaneously and support daisy-chain communication, allowing multiple monitoring devices to be connected together and supervised by a single microcontroller. This scalability makes modular BMS architectures particularly attractive for applications where battery configurations may vary between platforms.

Different communication architectures can be employed for connecting multiple AFE devices within a battery pack. One common approach is the daisy-chain architecture, where monitoring devices are connected sequentially through a communication link. This configuration reduces wiring complexity and simplifies system integration. However, communication failures in one monitoring device may affect communication with downstream devices. Alternatively, centralized architectures establish independent communication links between the microcontroller and each monitoring device. Although this approach increases wiring complexity and cost, it provides improved fault tolerance because the failure of one monitoring device does not interrupt communication with the remaining modules. To combine the advantages of both approaches, some battery systems employ dual-access daisy-chain communication, in which monitoring devices can be accessed from both ends of the communication chain. This architecture improves communication reliability while maintaining relatively low hardware complexity. The centralized and dual-access daisy-chain communication architectures are illustrated in Figures 2.4 and 2.5, respectively.

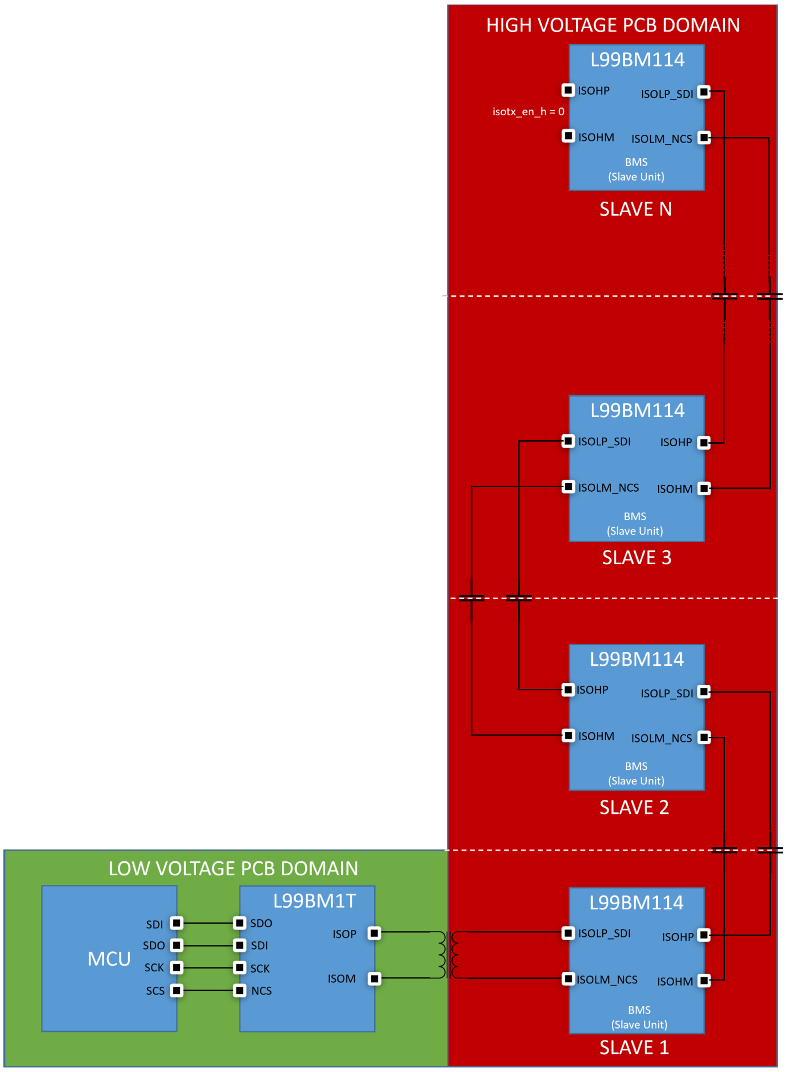

Figure 2.4 Single access BMS diagram.
Source: Reproduced from [10].

2.4.3 Cell Imbalance Phenomena

Cell imbalance is one of the most significant challenges encountered in battery packs containing multiple cells connected in series. Although cells may be manufactured according to identical specifications, variations in material properties, manufacturing tolerances, internal resistance, capacity, and self-discharge rates result in slightly different electrical behavior among cells. Consequently, cells within the same battery pack do not charge and discharge uniformly. Over time, these differences accumulate and lead to variations in the State of Charge (SOC) of individual cells.

Cell imbalance negatively affects battery performance, safety, and lifetime. During charging, cells with higher SOC may reach the maximum allowable voltage limit before the remaining cells are fully

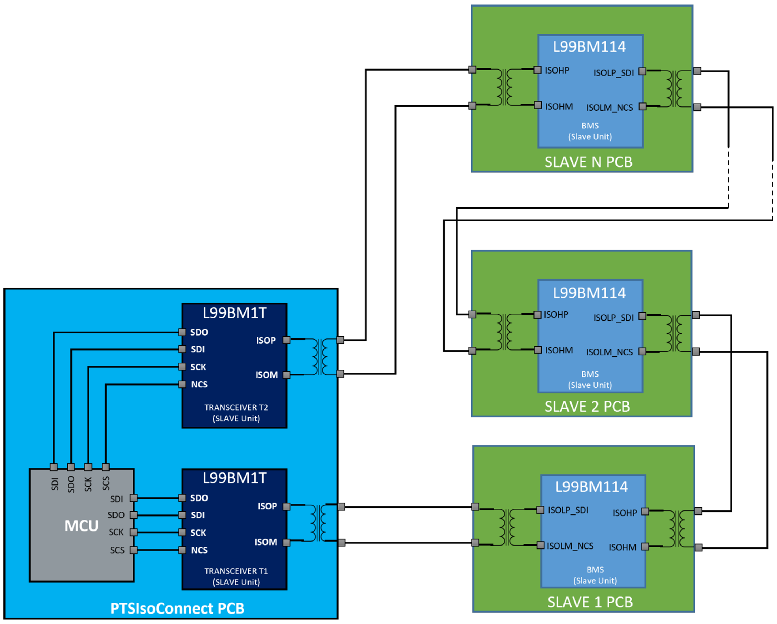

Figure 2.5 Dual access BMS diagram.
Source: Reproduced from [10].

charged. Similarly, during discharge, weaker cells may reach the minimum voltage limit earlier than other cells. As a result, the usable capacity of the entire battery pack becomes limited by the weakest cell. Furthermore, severe imbalance can accelerate battery degradation and increase heat generation within the pack. To mitigate these effects, battery balancing techniques are employed to maintain a more uniform charge distribution among cells.

Two primary balancing approaches are commonly used in Battery Management Systems: active balancing and passive balancing. Active balancing transfers energy from cells with higher SOC to cells with lower SOC using dedicated power electronic circuits. Since the energy is redistributed rather than dissipated, active balancing achieves high efficiency and minimizes energy losses. However, the implementation of active balancing requires additional hardware components and sophisticated control algorithms, resulting in increased system complexity and cost.

Passive balancing, in contrast, dissipates excess energy from highly charged cells through bleed resistors connected in parallel with the cells. When a cell reaches a higher SOC than neighboring cells, the balancing resistor is activated to discharge a portion of the stored energy until the cell voltage becomes comparable to the remaining cells. Although passive balancing results in energy losses in the form of heat, it is simpler to implement and requires fewer hardware resources. Consequently, passive balancing remains the most widely adopted balancing technique in commercial Battery Management Systems. The BMS architecture developed in this thesis utilizes passive balancing due to its simplicity, reliability, and suitability for the targeted application. Additional implementation details are presented in subsequent chapters[11].

2.4.4 Thermal Mnagement Considerations

Temperature is one of the most critical factors influencing the performance, safety, and lifetime of lithium-ion batteries. Battery operation outside the recommended temperature range can significantly reduce efficiency and accelerate degradation processes. Most lithium-ion batteries are designed to operate safely within a temperature range of approximately $-20\,°C$ to $60\,°C$, although the optimal operating range is typically much narrower. Elevated temperatures increase the rate of chemical side reactions within the cell, leading to accelerated capacity loss and increased internal resistance. In extreme cases, excessive temperatures may initiate thermal runaway, a self-sustaining process that can result in fire or explosion.

Low temperatures can also negatively affect battery performance. At reduced temperatures, the electrochemical reactions occurring inside the cell become less efficient, resulting in decreased power capability and reduced available capacity. Consequently, maintaining battery temperature within acceptable limits is essential for ensuring safe and reliable operation under varying environmental conditions.

The Battery Management System plays a central role in thermal monitoring and protection. Temperature measurements are commonly obtained using thermistors such as Negative Temperature Coefficient (NTC) sensors connected to the Analog Front-End device. The measured voltage values are converted into temperature information using calibration data provided by the sensor manufacturer. Based on the measured temperature, the BMS can generate warnings, trigger protective actions, or restrict battery operation when unsafe conditions are detected.

Several thermal management strategies can be employed to reduce battery temperatures and improve heat dissipation. Passive approaches include proper ventilation, the use of thermally conductive materials, and optimized battery packaging arrangements that facilitate heat transfer to the surrounding environment. Active cooling methods such as forced-air cooling and liquid cooling can provide improved thermal performance but increase system complexity, energy consumption, and weight. In unmanned aerial vehicle applications, where weight is a critical design constraint, passive cooling solutions are often preferred to minimize the impact on flight performance while maintaining acceptable thermal conditions for battery operation[12].

2.5 Battery Management Systems

2.5.1 Overview of Battery Management Systems

A Battery Management System (BMS) is a critical component in modern battery-powered systems, particularly in electric vehicles and unmanned aerial vehicles (UAVs). The primary role of the BMS is to ensure the safe, reliable, and efficient operation of the battery pack throughout its lifetime. In addition to monitoring battery parameters, the BMS acts as an interface between the battery pack and other subsystems, such as the flight controller, propulsion system, and charging system. By continuously evaluating battery conditions, the BMS can make operational decisions, issue warnings, and initiate protective actions when unsafe conditions are detected [13].

2.5.2 Functions of a Battery Management System

The functions of a Battery Management System can generally be divided into two categories: monitoring and control. The monitoring function involves continuously acquiring information about the battery pack, including cell voltages, current, and temperature. These measurements provide the information required to assess battery health and operating conditions. The control function utilizes the acquired data to regulate charging and discharging processes, determine operating limits, generate warning messages, and activate protection mechanisms when abnormal conditions occur. Through these functions, the BMS contributes significantly to battery safety, performance, and lifetime [1].

2.5.3 Battery Monitoring Functions

The Battery Management System continuously monitors three primary battery parameters: voltage, current, and temperature. These parameters provide the foundation for evaluating battery condition and determining whether the battery is operating within safe limits.

Cell voltage measurements are typically acquired directly by Analog Front-End (AFE) devices connected to the battery cells. The measured voltage values are transmitted to the microcontroller, where they are processed and compared against predefined safety thresholds. Voltage monitoring is essential for detecting overvoltage and undervoltage conditions, which can damage battery cells and reduce their operational lifetime.

Temperature monitoring is commonly performed using Negative Temperature Coefficient (NTC) thermistors placed in thermal contact with battery cells. The thermistor voltage is measured by the AFE and transmitted to the microcontroller. Using calibration data provided by the sensor manufacturer, the measured voltage can be converted into temperature values. Temperature monitoring enables the detection of excessive heating or operation in extremely cold environments, both of which can negatively affect battery performance and safety.

Battery current is typically measured using a precision shunt resistor connected in series with the battery pack. By measuring the voltage drop across the resistor and applying Ohm's law, the microcontroller can calculate the battery current. Current measurements are used to determine charging and discharging conditions and form the basis for higher-level calculations such as State of Charge (SOC) estimation and dynamic current limit determination.

The combination of voltage, current, and temperature measurements allows the BMS to estimate additional battery parameters, including State of Charge (SOC), available power capability, and safe charging and discharging current limits[1].

2.5.4 Protection Functions

One of the primary responsibilities of a Battery Management System is protecting the battery pack from unsafe operating conditions. Protection functions are implemented by continuously comparing measured battery parameters against predefined thresholds. When these thresholds are exceeded, the BMS generates warning flags or fault conditions and communicates them to higher-level controllers [1], [13].

Overvoltage Protection

Overvoltage occurs when a battery cell exceeds its maximum allowable voltage during charging. For most lithium-ion batteries, the maximum cell voltage is approximately 4.2 V. To prevent damage, the BMS may generate a warning when the cell voltage approaches this limit and declare a fault if the maximum allowable voltage is exceeded. Overvoltage protection is essential for preventing accelerated degradation and potential safety hazards [1], [13].

Undervoltage Protection

Undervoltage occurs when a battery cell is discharged below its minimum safe operating voltage. Although the exact threshold depends on cell chemistry and manufacturer specifications, lithium-ion cells are commonly protected from discharging below approximately 3.0 V. The BMS may generate a warning at a higher threshold, such as 3.4 V, and declare a fault when the minimum allowable voltage is reached. Undervoltage protection prevents irreversible battery damage and extends battery lifetime [1], [13].

Overtemperature Protection

High battery temperatures can accelerate degradation and increase the risk of thermal runaway. Therefore, the BMS continuously monitors battery temperature and generates warning and fault conditions when temperature thresholds are exceeded. For example, a warning may be issued at 50°C, while a fault condition may be triggered at a higher temperature limit. These thresholds are selected according to battery manufacturer specifications and application requirements [5], [12].

Undertemperature Protection

Operation at very low temperatures can significantly reduce battery performance and charging capability. To prevent battery damage, the BMS monitors cell temperature and restricts operation when temperatures fall below predefined limits. Warning and fault thresholds are typically established according to manufacturer recommendations and environmental operating conditions [5], [12].

Overcurrent Protection

Excessive charging or discharging current can generate excessive heat and accelerate battery degradation. Unlike voltage and temperature limits, allowable current limits are often dynamic and depend on several factors, including battery temperature, cell voltage, and manufacturer specifications. The BMS calculates the permissible charging and discharging currents and continuously compares them with measured current values. If the current exceeds the allowable limit, the system generates warnings or fault conditions to protect the battery pack [1], [13].

2.5.5 Battery Operating States

The Battery Management System operates using a state-based control strategy that determines the current operating mode of the battery pack. Typical operating states include Standby, Charging, Flight, and Fault. State transitions are determined by user requests and battery operating conditions. In the Standby state, the battery remains connected to the system but is neither charging nor supplying significant power. During Charging mode, the BMS supervises the charging process and continuously monitors battery parameters to ensure safe operation. In Flight mode, the battery supplies power to the propulsion system while the BMS continuously evaluates operating conditions and updates battery status information. If any protection threshold is violated, the BMS transitions to the Fault state, where appropriate protective actions are executed and fault information is communicated to the flight controller[13].
The transition between operating states is determined by both external requests and internal BMS logic. For example, a request to enter Flight mode may be rejected if an active fault condition exists. In such cases, the battery remains in the Fault state until the fault condition is cleared and safe operating conditions are restored[10], [13].

2.6 Modular Battery Management Systems

2.6.1 Centralized BMS Architecture

A centralized Battery Management System (BMS) architecture consists of a single control unit responsible for monitoring and managing all battery cells within the pack. In this configuration, voltage, current, and temperature measurements from every cell are routed to a central controller where all processing and decision-making functions are performed. Centralized architectures are commonly used in relatively small battery packs due to their simple control structure and low implementation cost.
Despite their simplicity, centralized BMS architectures become increasingly difficult to implement as battery pack size grows. The wiring harness required to connect every cell to a single controller becomes more complex, increasing system weight, assembly effort, and susceptibility to wiring faults. Furthermore, long analog signal paths can introduce measurement inaccuracies due to electromagnetic interference and noise. Consequently, centralized architectures are generally more suitable for battery systems containing a limited number of cells [14].

2.6.2 Distributed BMS Architecture

Distributed Battery Management Systems represent one of the most advanced battery monitoring architectures currently employed in large battery systems. In a distributed architecture, each battery module contains its own monitoring and control electronics. The local controllers are capable of processing measurements directly at the module level, significantly reducing the amount of analog wiring required throughout the battery pack.

One of the primary advantages of distributed architectures is their scalability. Additional battery modules can be incorporated into the system without requiring substantial modifications to the overall design. Furthermore, because measurements are processed locally, the impact of analog signal noise is significantly reduced. Distributed systems also improve modularity and simplify maintenance procedures. However, they generally require more hardware resources and can be more expensive than centralized solutions due to the increased number of controllers and monitoring devices [14].

2.6.3 Modular BMS Concepts

Modular Battery Management Systems are designed around the concept of dividing a battery pack into multiple independent modules. Each module is supervised by a Local Management Unit (LMU), while a Central Management Unit (CMU) coordinates the operation of the entire battery system. The LMUs are responsible for acquiring battery measurements such as cell voltages and temperatures, while the CMU performs higher-level management functions and system coordination.

This architecture offers several advantages, including improved scalability, maintainability, and fault tolerance. Since each module operates independently, faults occurring in one LMU do not necessarily affect the operation of the remaining modules. As a result, modular architectures are widely adopted in large battery systems where reliability and flexibility are critical requirements [15], [16].

2.6.4 Master–Slave Architecture

The master–slave architecture is one of the most common implementations of a modular Battery Management System. In this configuration, multiple slave boards are distributed throughout the battery pack and are responsible for monitoring cell voltages, temperatures, and other local measurements. The slave units collect and transmit measurement data to a master controller through a communication interface.

Unlike the slave units, the master controller performs all system-level processing and decision-making tasks. It receives data from all slave boards, executes battery management algorithms, determines operating states, and generates warning or fault conditions when necessary. This architecture provides a clean and scalable solution for monitoring large battery packs containing a high number of series-connected cells.

The master–slave architecture offers several advantages, including simplified system expansion, centralized control logic, and improved electrical isolation between measurement circuits and the main controller. However, it also introduces a single point of failure, since the loss of the master controller can result in the loss of monitoring and control capabilities for the entire battery system [16].

2.6.5 Analog Front-End Devices for Battery Monitoring

Analog Front-End (AFE) devices are specialized integrated circuits designed for battery monitoring applications. Their primary function is to acquire analog measurements from battery cells and convert them into digital information that can be processed by a microcontroller. Typical measurements include cell voltages, temperature sensor readings, and balancing status information.

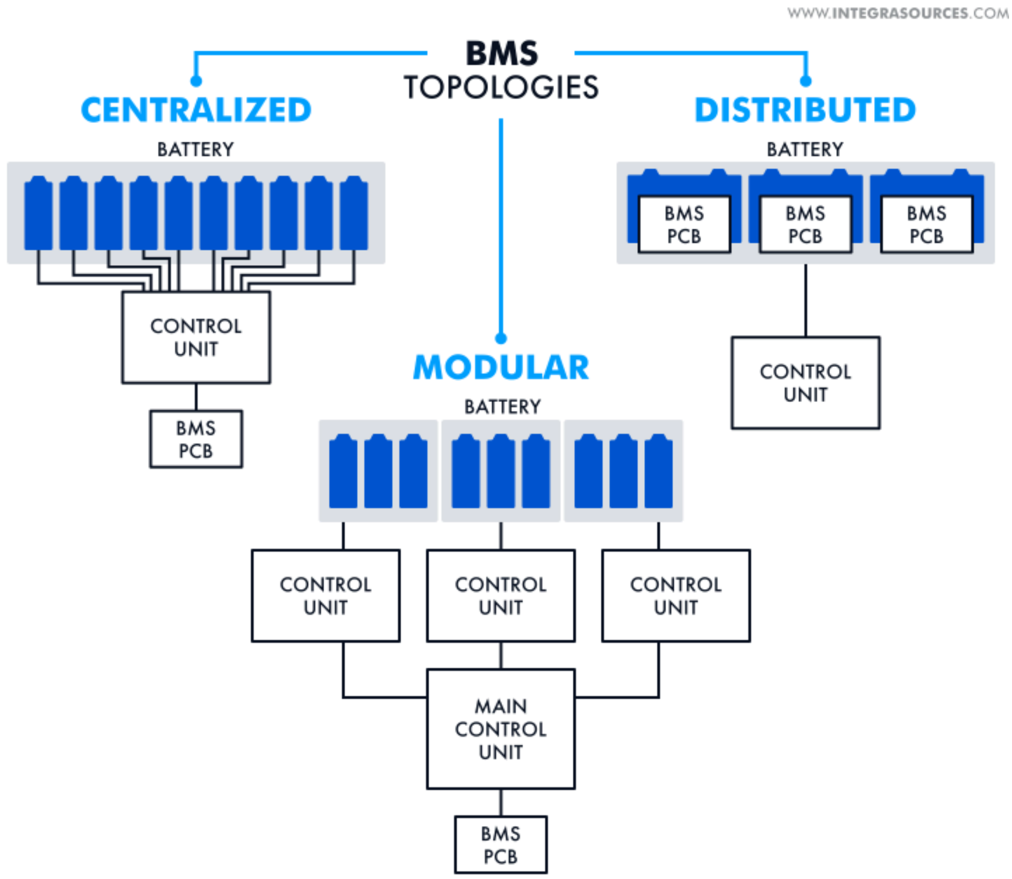

Figure 2.6 Different BMS archirectures.
Source: Adapted from Integra Sources [17].

The selection of an AFE device depends on several factors, including the number of cells to be monitored, measurement accuracy requirements, communication interfaces, and system cost constraints. Modern AFE devices are capable of monitoring multiple battery cells simultaneously and often include integrated protection and balancing features.

In the proposed battery management system developed in this thesis, the selected AFE device is capable of monitoring up to fourteen series-connected cells. Multiple AFE devices can be interconnected to support larger battery packs while maintaining a modular architecture. Communication between the AFE devices and the microcontroller can be achieved using various communication protocols depending on system requirements. The measured data is transmitted to the microcontroller, where battery management algorithms process the information and determine the appropriate control actions [10].

2.7 Embedded Systems for Battery Management

2.7.1 Microcontroller-Based BMS Implementations

The microcontroller represents the central processing unit of a Battery Management System and is responsible for acquiring measurements, executing control algorithms, and making operational decisions. The selection of an appropriate microcontroller is therefore a critical design step in any

BMS implementation. Factors influencing the selection process include computational requirements, memory resources, communication interfaces, power consumption, and compatibility with monitoring devices such as Analog Front-End (AFE) integrated circuits. In addition to hardware capabilities, the software development environment associated with the microcontroller should also be considered, as it directly affects development efficiency, debugging, testing, and long-term maintainability of the system [15].

2.7.2 STM32 Microcontroller Family

The STM32 family of microcontrollers, developed by STMicroelectronics, has become widely adopted in embedded systems due to its performance, flexibility, and extensive peripheral support. These microcontrollers provide a variety of communication interfaces, including UART, CAN, SPI, and I^2C, making them suitable for battery management applications. Furthermore, STM32 devices offer sufficient processing power and memory resources to execute battery monitoring, protection, balancing, and communication tasks simultaneously. In this thesis, an STM32 microcontroller is employed as the central controller of the BMS due to its compatibility with Model-Based Design workflows and its ability to efficiently process battery measurements while maintaining reliable communication with external devices [18].

2.7.3 Communication Interfaces in BMS

Communication interfaces play a vital role in Battery Management Systems by enabling data exchange between monitoring devices, microcontrollers, and external subsystems. Modern BMS architectures often utilize multiple communication protocols simultaneously to satisfy different system requirements. The selected communication interface must provide sufficient speed, reliability, and noise immunity while maintaining compatibility with the hardware architecture of the battery system [18].

CAN Communication

The Controller Area Network (CAN) protocol is widely used in automotive and aerospace applications due to its robustness and fault tolerance. CAN enables reliable communication between multiple electronic control units while maintaining immunity to electrical noise. In battery-powered vehicles and drones, CAN is commonly employed to exchange battery status information with higher-level controllers. Although CAN communication is not used for battery monitoring within the proposed BMS architecture, it can be utilized for communication between the battery system and other embedded subsystems, such as the flight controller or power management units [19].

UART Communication

Universal Asynchronous Receiver Transmitter (UART) communication provides a simple and effective method for serial data transfer between electronic devices. UART is commonly used for debugging, data logging, firmware updates, and communication with personal computers. In the proposed system, UART communication is utilized for interaction between the STM32 microcontroller and the host computer during software development, code deployment, and system testing. The simplicity of

UART communication makes it particularly useful during the development and validation phases of embedded systems [18].

SPI and isoSPI Communication

The Serial Peripheral Interface (SPI) protocol is a high-speed synchronous communication interface commonly used for communication between microcontrollers and peripheral devices. A standard SPI interface utilizes four signals: Serial Clock (SCK), Master Out Slave In (MOSI), Master In Slave Out (MISO), and Chip Select (CS). In Battery Management Systems, SPI is frequently employed to communicate with monitoring devices such as Analog Front-End integrated circuits.
For battery packs containing multiple series-connected cells, communication reliability becomes increasingly important due to the presence of high common-mode voltages and electrical noise. To address these challenges, some AFE devices employ isolated SPI communication, commonly referred to as isoSPI. Instead of transmitting data using single-ended signals, isoSPI utilizes differential signaling through two communication lines. Since information is transmitted as the voltage difference between the two conductors, the communication link exhibits significantly improved noise immunity. In such systems, dedicated transceiver devices are used to convert conventional SPI signals generated by the microcontroller into isoSPI signals and vice versa, enabling reliable communication between the microcontroller and the battery monitoring devices [10].

2.8 Model-Based Design for Embedded Systems

2.8.1 Principles of Model-Based Design

Model-Based Design (MBD) is a development methodology that utilizes graphical system models to design, analyze, verify, and implement complex embedded systems. Rather than beginning with software implementation directly, engineers first construct a mathematical and functional representation of the system using simulation tools. Input signals, control logic, and expected outputs are evaluated through simulation before hardware deployment. This approach enables design issues to be identified and corrected early in the development process, thereby reducing development costs and improving system reliability. In battery management applications, Model-Based Design facilitates the rapid development and verification of monitoring, protection, and control algorithms before deployment on the target hardware platform.

2.8.2 MATLAB/Simulink Environment

MATLAB and Simulink provide one of the most widely used Model-Based Design environments for embedded system development. Simulink offers an extensive library of pre-built blocks that can be used to model physical systems, communication interfaces, control algorithms, and embedded software. In addition, specialized libraries are available for numerous microcontroller platforms, including STM32 devices. These libraries enable direct interaction with hardware peripherals and facilitate automatic code generation. The MATLAB ecosystem also provides numerous additional tools that support battery system modeling, data analysis, testing, and verification, many of which are utilized throughout this thesis [20].

2.8.3 Stateflow for State-Machine Development

Stateflow is a graphical modeling environment integrated within Simulink that enables the development of state machines and event-driven control systems. Battery Management Systems frequently rely on state-based logic to determine operating modes and protection actions. Examples include transitions between standby, charging, discharging, and fault states. Stateflow allows these operational modes to be represented visually as interconnected states linked by transition conditions. When a specified condition becomes true, the system transitions from one state to another and executes the associated actions. This graphical representation improves readability, simplifies debugging, and provides a clear description of system behavior [21].

2.8.4 Automatic Code Generation

One of the most significant advantages of Model-Based Design is the ability to automatically generate production-ready source code directly from the developed model. Rather than manually implementing algorithms in a conventional programming environment, the engineer develops and verifies the functionality within Simulink and then generates the corresponding embedded software automatically. This process reduces development time, minimizes programming errors, and ensures consistency between the simulated model and the deployed implementation. For STM32-based systems, generated code can be integrated directly into the microcontroller development workflow, significantly simplifying software deployment [22].

2.8.5 Advantages of Model-Based Design in BMS Development

Model-Based Design offers several advantages for Battery Management System development. The availability of extensive software libraries, reference models, and technical documentation significantly reduces development effort. Furthermore, graphical interfaces allow system behavior to be visualized during simulation, making it easier to understand complex interactions between subsystems. Simulink dashboard components can display system status information through indicators, gauges, and warning signals, providing immediate feedback during testing. Additional tools such as Test Sequence and Test Harness environments enable automated testing of individual subsystems under a variety of operating scenarios. These capabilities contribute to improved design quality, increased productivity, and more comprehensive validation of battery management algorithms [21].

2.9 Verification and Validation Methods

2.9.1 Verification and Validation in Embedded Systems

Verification and Validation (V&V) activities are essential for ensuring the reliability and safety of embedded systems. Verification focuses on confirming that the implemented system satisfies its specified requirements, while validation ensures that the system fulfills its intended operational purpose. Within a Model-Based Design workflow, verification and validation are performed at multiple stages of development through progressively more realistic testing environments. This structured approach enables design flaws to be identified early and reduces the risk of failures during hardware deployment [23].

2.9.2 Model-in-the-Loop (MIL)

Model-in-the-Loop testing represents the first stage of the verification process. In this environment, all system components are implemented using Simulink models and mathematical representations. Control algorithms, protection logic, and system responses are evaluated entirely within the simulation environment. MIL testing provides a rapid method for validating functional requirements and confirming correct system behavior before software code generation [21], [23].

2.9.3 Software-in-the-Loop (SIL)

Software-in-the-Loop testing evaluates the software implementation generated from the simulation model. During SIL testing, automatically generated source code is executed on the host computer and compared against the original model behavior. This process verifies that the generated code accurately reproduces the functionality defined in the simulation model and helps identify discrepancies introduced during code generation [24].

2.9.4 Processor-in-the-Loop (PIL)

Processor-in-the-Loop testing extends the verification process by executing the generated code directly on the target microcontroller. In this configuration, the software runs on the actual STM32 hardware while remaining connected to the simulation environment. PIL testing verifies correct integration between the generated software and the target processor and allows performance characteristics such as execution time and memory usage to be evaluated [25].

2.9.5 Hardware-in-the-Loop (HIL)

Hardware-in-the-Loop testing represents the final stage of the verification and validation process. In a HIL environment, the physical hardware interacts with a real-time simulation that emulates the behavior of the battery system and surrounding environment. This approach enables the embedded controller to be tested under realistic operating conditions without exposing actual hardware to unnecessary risk. For a well-designed system, the results obtained from HIL testing should closely match those observed during MIL, SIL, and PIL verification stages, thereby providing confidence in the overall design and implementation [26].

2.10 Research Gap

Existing research has extensively investigated Battery Management Systems, modular battery architectures, and Model-Based Design methodologies independently. Significant progress has also been achieved in battery monitoring, fault detection, and embedded control implementation. However, limited research has focused on integrating a scalable modular BMS architecture, AFE-based battery monitoring, automatic code generation, and a complete Model-in-the-Loop, Software-in-the-Loop, Processor-in-the-Loop, and Hardware-in-the-Loop validation workflow within a unified development framework.

Furthermore, emerging research trends are exploring the integration of artificial intelligence techniques for advanced battery diagnostics and wireless communication technologies for next-generation Battery Management Systems. Although these approaches offer promising opportunities for future developments, challenges related to reliability, computational requirements, and safety certification remain active research topics.

Therefore, there remains a need for a flexible and scalable BMS development framework that combines modular hardware architecture, Model-Based Design principles, automated code generation, and comprehensive verification and validation procedures. This thesis aims to address this gap through the development and validation of a modular Battery Management System for UAV applications [13].

A Appendix

For later

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